

First order logic

Introduction, syntax and semantics

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Expressivity of propositional logic - I

Question

Try to express in Propositional Logic the following statements:

- Mary is a person
- John is a person
- Mary is mortal
- Mary and John are siblings

A solution

Through atomic propositions:

- Mary-is-a-person
- John-is-a-person
- Mary-is-mortal
- Mary-and-John-are-siblings

Problem with previous solution

- Mary-is-a-person
- John-is-a-person
- Mary-is-mortal
- Mary-and-John-are-siblings

How do we link Mary of the first sentence to Mary of the third sentence? Same with John. How do we link Mary and Mary-and-John?

Expressivity of propositional logic - II

Question

Try to express in Propositional Logic the following statements:

- All persons are mortal;
- There is a person who is a spy.

A solution

We can give all people a name and express this fact through atomic propositions:

- $\text{Mary-is-mortal} \wedge \text{John-is-mortal} \wedge \text{Chris-is-mortal} \wedge \dots \wedge \text{Michael-is-mortal}$
- $\text{Mary-is-a-spy} \vee \text{John-is-a-spy} \vee \text{Chris-is-a-spy} \vee \dots \vee \text{Michael-is-a-spy}$

Problem with previous solution

- $\text{Mary-is-mortal} \wedge \text{John-is-mortal} \wedge \text{Chris-is-mortal} \wedge \dots \wedge \text{Michael-is-mortal}$
- $\text{Mary-is-a-spy} \vee \text{John-is-a-spy} \vee \text{Chris-is-a-spy} \vee \dots \vee \text{Michael-is-a-spy}$

The representation is not compact and generalization patterns are difficult to express.

What if we do not know all the people in our “universe”? How can we express the statement independently from the people in our “universe”?

Expressivity of propositional logic - III

Question

Try to express in Propositional Logic the following statements:

- Every natural number is either even or odd

A solution

We can use two families of propositions $even_i$ and odd_i for every $i \geq 1$, and use the set of formulas

$$\{odd_i \vee even_i | i \geq 1\}$$

Problem with previous solution

$$\{odd_i \vee even_i | i \geq 1\}$$

What happens if we want to state this in one single formula? To do this we would need to write an infinite formula like:

$$(odd_1 \vee even_1) \wedge (odd_2 \vee even_2) \wedge \dots$$

and this cannot be done in propositional logic.

Expressivity of propositional logic -IV

Question

Express the statements:

- the father of Luca is Italian

Solution (Partial)

- `mario-is-father-of-luca` $\rightarrow$ `mario-is-italian`
- `michele-is-father-of-luca` $\rightarrow$ `michele-is-italian`
- ...

Problem with previous solution

- `mario-is-father-of-luca` $\rightarrow$ `mario-is-italian`
- `michele-is-father-of-luca` $\rightarrow$ `michele-is-italian`
- ...

This statement strictly depend from a fixed set of people. What happens if we want to make this statement independently of the set of persons we have in our universe?

Why first order logic?

Because it provides a way of **representing** information like the following one:

- ① **Mary is a person;**
- ② John is a person;
- ③ **Mary is mortal;**
- ④ Mary and John are siblings
- ⑤ **Every person is mortal;**
- ⑥ There is a person who is a spy;
- ⑦ Every natural number is either even or odd;
- ⑧ The father of Luca is Italian

and also to **infer** the third one from the first one and the fifth one.

First order logic

While propositional logic assumes that the world to be formalized is constituted by facts corresponding to propositions, first-order logic (like natural language) assumes that the world is constituted by:

- **Individual objectes**, denoted by **Constants**: mary, john, 1, 2, 3, red, blue, world war 1, world war 2, 18th Century. . .
- **Functional means to refer to objects**, denoted by **Functions**: the father of, the best friend, the third inning of, . . .
- **Properties and relations**, denoted by **Predicates**: Mortal, Prime, IsBrotherOf, Bigger than, Inside, IsPartOf, HasColor, Owns, . . .

In the following, we define first-order logic **with equality**.

Constants and Predicates

- Mary is a person
- John is a person
- Mary is mortal
- Mary and John are siblings

In FOL it is possible to build an atomic propositions by applying a **predicate** to **constants**

- *Person(mary)*
- *Person(john)*
- *Mortal(mary)*
- *Siblings(mary, john)*

Quantifiers and variables

- Every person is mortal;
- There is a person who is a spy;
- Every natural number is either even or odd;

In FOL it is possible to build propositions by applying **universal** (**existential**) **quantifiers** to **variables**. This allows to quantify to arbitrary objects of the universe.

- $\forall x. Person(x) \rightarrow Mortal(x)$
- $\exists x. Person(x) \wedge Spy(x)$
- $\forall x. (Odd(x) \vee Even(x))$

- The father of Luca is Italian.

In FOL it is possible to build propositions by applying a **function** to a **constant**, and then a predicate to the resulting object.

- *Italian(fatherOf(Luca))*

Syntax of FOL

The **alphabet of FOL** is composed of two sets of symbols:

Logical symbols

- the logical constants $\perp$ (false) and $\top$ (true)
- propositional logical connectives $\wedge, \vee, \rightarrow, \neg, \equiv$
- the **quantifiers** $\forall, \exists$
- a denumerable set of **variable symbols** $x_1, x_2, \dots$
- the **equality predicate symbol** $=$ (optional)

Non logical symbols $\langle c_1, c_2, \dots, f_1, f_2, \dots, P_1, P_2, \dots \rangle$

- a denumerable set $c_1, c_2, \dots$ of **constants**
- a denumerable set $f_1, f_2, \dots$ of **function symbols** each of which is associated with its **arity** ≥ 1 (i.e., number of arguments)
- a denumerable set $P_1, P_2, \dots$ of **predicate (or relational) symbols** each of which is associated with its **arity** ≥ 0 (i.e., number of arguments)